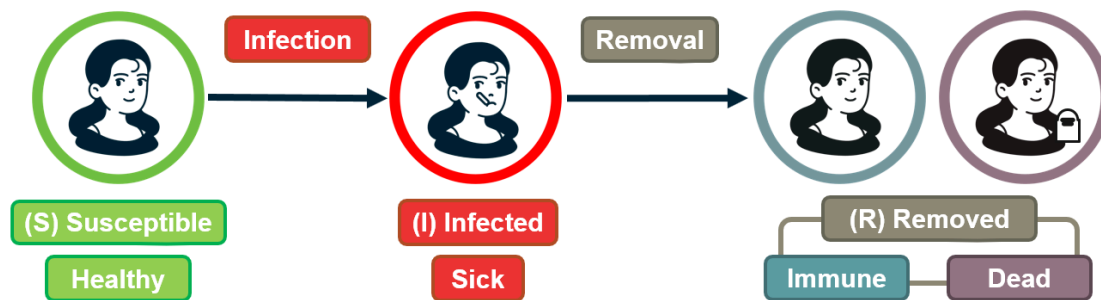


## L9: SIR Model



For some pathogens (e.g., the virus VZV that causes chickenpox), if an individual recovers he/she develops persistent immunity (through the creation of antibodies for that pathogen) and so the individual cannot get infected again.

For other pathogens, such as HIV, there is no natural recovery and an infected individual may die after some time.

To model both possibilities, the SIR model extends the SI model with a third state R referred to as “**Removed**”. The transition from I to R represents that either the infected individual acquired natural immunity or that he/she died. In either case, that individual cannot infect anyone else and cannot get infected again.

As in the case of the SIS model, we will denote as  $\mu$  the parameter that describes how fast an infected individual moves out of the infected state (independent of whether this transition represents recovery/immunity or death).

There are now three population densities, one for each state, and they should always add up to one:  $s(t) + i(t) + r(t) = 1$ .

(To the right: Figure 10.6 from Network Science by Albert-László Barabási)

Similar with the SIS model, we can write a differential equation for the density of infected individuals:

$$\frac{di(t)}{dt} = \bar{k}\beta i(t)s(t) - \mu i(t)$$

The only difference with SIR is that  $s(t) = 1 - i(t) - r(t)$ .

The differential equation for the density of removed individuals is simply:

$$\frac{dr(t)}{dt} = \mu i(t)$$

At this point we have a system of two differential equations, for  $i(t)$  and  $r(t)$ , with the initial conditions:  $i(0) = i_0$  and  $r(0) = 0$ .

If we solve these equations, the density of S individuals is simply  $s(t) = 1 - i(t) - r(t)$ .

The previous system of differential equations cannot be solved analytically, however. Numerically, we get plots such as the visualization (for the case  $\bar{k}\beta > \mu$ ). In this case, the initial outbreak leads to an epidemic in which all the individuals first move to the infected state (green curve) and then to the removed state (purple line).

If  $\bar{k}\beta < \mu$ , the initial outbreak dies out as in the case of the SIS model, and almost the entire population remains in the S state.

So the epidemic threshold for the SIR model is also equal to one, as in the case of the SIS model.

